2.27 Integration: Inverse Hyperbolic Functions

Worked Solutions

1)

Let $x = \sinh u$, so $dx = \cosh u\, du$. Then

$$\begin{aligned} I &= \int \frac{\cosh u}{\sqrt{\sinh^2 u + 1}}\, du = \int \frac{\cosh u}{\sqrt{\cosh^2 u}}\, du = \int 1\, du \\ &= u + c = \operatorname{arsinh} x + c. \end{aligned}$$

2)

Let $x = \cosh u$, so $dx = \sinh u\, du$. Then

$$\begin{aligned} I &= \int \frac{\sinh u}{\sqrt{\cosh^2 u - 1}}\, du = \int \frac{\sinh u}{\sqrt{\sinh^2 u}}\, du = \int 1\, du \\ &= u + c = \operatorname{arcosh} x + c. \end{aligned}$$

3)

Let $x = \tanh u$, so $dx = \operatorname{sech}^2 u\, du$. Since $1 - \tanh^2 u = \operatorname{sech}^2 u$,

$$I = \int \frac{\operatorname{sech}^2 u}{1 - \tanh^2 u}\, du = \int 1\, du = u + c = \operatorname{artanh} x + c.$$

4 a)

$$\int \frac{4}{\sqrt{x^2 - 1}}\, dx = 4 \int \frac{1}{\sqrt{x^2 - 1}}\, dx = 4 \operatorname{arcosh} x + c.$$

4 b)

$$\int \frac{1}{2\sqrt{x^2 + 1}}\, dx = \frac{1}{2} \int \frac{1}{\sqrt{x^2 + 1}}\, dx = \frac{1}{2} \operatorname{arsinh} x + c.$$

4 c)

$$\int \frac{6}{1 - x^2}\, dx = 6 \int \frac{1}{1 - x^2}\, dx = 6 \operatorname{artanh} x + c.$$

4 d)

$$\int \frac{1}{\sqrt{4x^2 - 1}}\, dx = \int \frac{1}{\sqrt{(2x)^2 - 1}}\, dx = \frac{1}{2} \operatorname{arcosh}(2x) + c.$$

4 e)

$$\int \frac{1}{\sqrt{9x^2+1}}\,dx = \int \frac{1}{\sqrt{(3x)^2+1}}\,dx = \frac{1}{3}\operatorname{arsinh}(3x) + c.$$

4 f)

$$\int \frac{1}{1-25x^2}\,dx = \int \frac{1}{1-(5x)^2}\,dx = \frac{1}{5}\operatorname{artanh}(5x) + c.$$

4 g)

$$\int \frac{10}{\sqrt{16x^2-1}}\,dx = 10\int \frac{1}{\sqrt{(4x)^2-1}}\,dx = \frac{10}{4}\operatorname{arcosh}(4x) + c = \frac{5}{2}\operatorname{arcosh}(4x) + c.$$

4 h)

$$\begin{aligned}\int \frac{\sqrt{8}}{\sqrt{2x^2+1}}\,dx &= 2\sqrt{2}\int \frac{1}{\sqrt{(\sqrt{2}x)^2+1}}\,dx \\ &= 2\sqrt{2}\cdot\frac{1}{\sqrt{2}}\operatorname{arsinh}(\sqrt{2}x) + c = 2\operatorname{arsinh}(\sqrt{2}x) + c.\end{aligned}$$

4 i)

$$\int \frac{100}{1-\frac{1}{5}x^2}\,dx = 100\sqrt{5}\,\operatorname{artanh}\left(\frac{x}{\sqrt{5}}\right) + c = 100\sqrt{5}\,\operatorname{artanh}\left(\frac{\sqrt{5}}{5}x\right) + c.$$

5 a)

$$\frac{1}{15-x^2} = \frac{1}{15\left(1-\frac{1}{15}x^2\right)} = \frac{1}{15\left(1-\left(\frac{x}{\sqrt{15}}\right)^2\right)}.$$

Thus $k = 1/\sqrt{15} = \sqrt{15}/15$.

5 b)

$$\begin{aligned}\int \frac{1}{15-x^2}\,dx &= \frac{1}{15}\int \frac{1}{1-(x/\sqrt{15})^2}\,dx \\ &= \frac{1}{15}\cdot\sqrt{15}\,\operatorname{artanh}\left(\frac{x}{\sqrt{15}}\right) + c = \frac{\sqrt{15}}{15}\operatorname{artanh}\left(\frac{\sqrt{15}}{15}x\right) + c.\end{aligned}$$

6 a)

$$\frac{1}{\sqrt{x^2+3}} = \frac{1}{\sqrt{3}\sqrt{\frac{1}{3}x^2+1}} = \frac{1}{\sqrt{3}\sqrt{\left(\frac{x}{\sqrt{3}}\right)^2+1}}.$$

Thus $k = 1/\sqrt{3} = \sqrt{3}/3$.

6 b)

$$\begin{aligned}\int \frac{1}{\sqrt{x^2+3}}\,dx &= \frac{1}{\sqrt{3}}\int \frac{1}{\sqrt{(x/\sqrt{3})^2+1}}\,dx \\ &= \operatorname{arsinh}\left(\frac{x}{\sqrt{3}}\right)+c = \operatorname{arsinh}\left(\frac{\sqrt{3}}{3}x\right)+c.\end{aligned}$$

7 a)

$$\int \frac{1}{25-x^2}\,dx = \frac{1}{5}\int \frac{1}{1-(x/5)^2}\,dx = \frac{1}{5}\operatorname{artanh}\left(\frac{x}{5}\right)+c.$$

7 b)

$$\int \frac{1}{\sqrt{x^2+9}}\,dx = \frac{1}{3}\int \frac{1}{\sqrt{(x/3)^2+1}}\,dx = \operatorname{arsinh}\left(\frac{x}{3}\right)+c.$$

7 c)

$$\int \frac{1}{\sqrt{x^2-4}}\,dx = \frac{1}{2}\int \frac{1}{\sqrt{(x/2)^2-1}}\,dx = \operatorname{arcosh}\left(\frac{x}{2}\right)+c.$$

7 d)

$$\int \frac{1}{8-x^2}\,dx = \frac{1}{8}\int \frac{1}{1-(x/\sqrt{8})^2}\,dx = \frac{\sqrt{2}}{4}\operatorname{artanh}\left(\frac{\sqrt{2}}{4}x\right)+c.$$

7 e)

$$\begin{aligned}\int \frac{1}{\sqrt{x^2+12}}\,dx &= \frac{1}{\sqrt{12}}\int \frac{1}{\sqrt{(x/\sqrt{12})^2+1}}\,dx \\ &= \operatorname{arsinh}\left(\frac{x}{\sqrt{12}}\right)+c = \operatorname{arsinh}\left(\frac{\sqrt{3}}{6}x\right)+c.\end{aligned}$$

7 f)

$$\int \frac{1}{\sqrt{x^2-20}}\,dx = \frac{1}{\sqrt{20}}\int \frac{1}{\sqrt{(x/\sqrt{20})^2-1}}\,dx$$
$$= \operatorname{arcosh}\left(\frac{x}{\sqrt{20}}\right)+c = \operatorname{arcosh}\left(\frac{\sqrt{5}}{10}x\right)+c.$$

7 g)

$$\int \frac{5}{10-x^2}\,dx = \frac{1}{2}\int \frac{1}{1-(x/\sqrt{10})^2}\,dx = \frac{\sqrt{10}}{2}\operatorname{artanh}\left(\frac{x}{\sqrt{10}}\right)+c.$$

7 h)

$$\int \frac{14}{\sqrt{x^2+7}}\,dx = \frac{14}{\sqrt{7}}\int \frac{1}{\sqrt{(x/\sqrt{7})^2+1}}\,dx = 14\operatorname{arsinh}\left(\frac{x}{\sqrt{7}}\right)+c.$$

7 i)

$$\int \frac{18}{\sqrt{x^2-3}}\,dx = \frac{18}{\sqrt{3}}\int \frac{1}{\sqrt{(x/\sqrt{3})^2-1}}\,dx = 18\operatorname{arcosh}\left(\frac{x}{\sqrt{3}}\right)+c.$$

7 j)

$$\int \frac{1}{9-4x^2}\,dx = \frac{1}{9}\int \frac{1}{1-(2x/3)^2}\,dx = \frac{1}{6}\operatorname{artanh}\left(\frac{2}{3}x\right)+c.$$

7 k)

$$\int \frac{1}{\sqrt{9x^2+4}}\,dx = \frac{1}{2}\int \frac{1}{\sqrt{(3x/2)^2+1}}\,dx = \frac{1}{3}\operatorname{arsinh}\left(\frac{3}{2}x\right)+c.$$

7 l)

$$\int \frac{1}{\sqrt{25x^2-16}}\,dx = \frac{1}{4}\int \frac{1}{\sqrt{(5x/4)^2-1}}\,dx = \frac{1}{5}\operatorname{arcosh}\left(\frac{5}{4}x\right)+c.$$

7 m)

$$\int \frac{15}{4-\frac{1}{4}x^2}\,dx = \frac{15}{4}\int \frac{1}{1-(x/4)^2}\,dx = 15\,\operatorname{artanh}\left(\frac{x}{4}\right)+c.$$

7 n)

$$\int \frac{3}{\sqrt{16x^2+\frac{1}{9}}}\,dx = 9\int \frac{1}{\sqrt{(12x)^2+1}}\,dx = \frac{3}{4}\operatorname{arsinh}(12x)+c.$$

7 o)

$$\int \frac{18}{\sqrt{\frac{4}{9}x^2-\frac{9}{4}}}\,dx = 108\int \frac{1}{\sqrt{(4x)^2-9^2}}\,dx = 27\operatorname{arcosh}\left(\frac{4x}{9}\right)+c.$$

8 a)

$$\int \frac{1}{a^2-x^2}\,dx = \frac{1}{a^2}\int \frac{1}{1-(x/a)^2}\,dx = \frac{1}{a}\operatorname{artanh}\left(\frac{x}{a}\right)+c.$$

8 b)

$$\int \frac{1}{\sqrt{x^2+a^2}}\,dx = \frac{1}{a}\int \frac{1}{\sqrt{(x/a)^2+1}}\,dx = \operatorname{arsinh}\left(\frac{x}{a}\right)+c.$$

8 c)

$$\int \frac{1}{\sqrt{k^2x^2-a^2}}\,dx = \frac{1}{a}\int \frac{1}{\sqrt{(kx/a)^2-1}}\,dx = \frac{1}{k}\operatorname{arcosh}\left(\frac{kx}{a}\right)+c.$$

9 a)

$$\int_{-1/2}^{1/2} \frac{1}{1-x^2}\,dx = \left[\operatorname{artanh} x\right]_{-1/2}^{1/2} = \operatorname{artanh}\frac{1}{2} - \operatorname{artanh}\left(-\frac{1}{2}\right) = \ln 3.$$

9 b)

$$\int_1^2 \frac{1}{\sqrt{4x^2-1}}\,dx = \left[\frac{1}{2}\operatorname{arcosh}(2x)\right]_1^2 = \frac{1}{2}\ln\left(\frac{4+\sqrt{15}}{2+\sqrt{3}}\right).$$

9 c)

$$\int_{-5}^{5} \frac{30}{\sqrt{36x^2+25}}\,dx = \left[5\,\mathrm{arsinh}\left(\frac{6x}{5}\right)\right]_{-5}^{5} = 10\,\mathrm{arsinh}\,6 = 10\ln(6+\sqrt{37}).$$

10)

The shaded region is between $x=-1$ and $x=1$ and the curve, so

$$\begin{aligned} A &= \int_{-1}^{1} \frac{3}{9-4x^2}\,dx = \left[\frac{1}{2}\,\mathrm{artanh}\left(\frac{2x}{3}\right)\right]_{-1}^{1} \\ &= \mathrm{artanh}\,\frac{2}{3} = \frac{1}{2}\ln\left(\frac{1+2/3}{1-2/3}\right) = \frac{1}{2}\ln 5. \end{aligned}$$

11)

$$\begin{aligned} &\int_{3/5}^{4/5} \left(\frac{5}{\sqrt{25x^2-4}} + \frac{31}{155x-92}\right)dx \\ &= \left[\mathrm{arcosh}\left(\frac{5}{2}x\right) + \frac{1}{5}\ln|155x-92|\right]_{3/5}^{4/5} \\ &= \mathrm{arcosh}\,2 + \frac{1}{5}\ln 32 - \mathrm{arcosh}\,\frac{3}{2} - \frac{1}{5}\ln 1 \\ &= \ln(2+\sqrt{3}) + \ln 2 - \ln\left(\frac{3}{2}+\frac{\sqrt{5}}{2}\right) \\ &= \ln\left(\frac{8+4\sqrt{3}}{3+\sqrt{5}}\right) = \ln(6-2\sqrt{5}+3\sqrt{3}-\sqrt{15}). \end{aligned}$$

12)

$$\begin{aligned} \int_{1}^{3/2} \frac{4}{\sqrt{4x^2-1}}\,dx &= \left[2\,\mathrm{arcosh}(2x)\right]_1^{3/2} \\ &= 2\ln(3+2\sqrt{2}) - 2\ln(2+\sqrt{3}) \\ &= \ln\left(\frac{(3+2\sqrt{2})^2}{(2+\sqrt{3})^2}\right) = \ln\left(\frac{17+12\sqrt{2}}{7+4\sqrt{3}}\right) \\ &= \ln(119-48\sqrt{6}-68\sqrt{3}+84\sqrt{2}). \end{aligned}$$

Thus $a=119$, $b=-48$, $c=-68$ and $d=84$.

13 a)

$$19-39x^2 \equiv (Ax+B)(4-9x^2) + (Cx+D)(1-x^2).$$

At $x=1$, $-20=-5(A+B)$, so $A+B=4$. At $x=-1$, $-20=-5(-A+B)$, so $-A+B=4$. At $x=2/3$, $5/3=(2C/3+D)(5/9)$, so $2C/3+D=3$. At $x=-2/3$, $5/3=(-2C/3+D)(5/9)$, so $-2C/3+D=3$. Therefore $A=C=0$, $B=4$ and $D=3$, giving

$$f(x) = \frac{4}{1-x^2} + \frac{3}{4-9x^2}.$$

13 b)

$$\begin{aligned}\int f(x)\,dx &= \int \left(\frac{4}{1-x^2} + \frac{3}{4-9x^2}\right) dx \\ &= 4\operatorname{artanh} x + \frac{3}{4}\int \frac{1}{1-(3x/2)^2}\,dx \\ &= 4\operatorname{artanh} x + \frac{1}{2}\operatorname{artanh}\left(\frac{3}{2}x\right) + c.\end{aligned}$$

Hence $p = 4$, $q = 1$, $r = 1/2$ and $s = 3/2$.

14 a)

Write

$$\frac{12x^2+1}{(1-9x^2)(1-16x^2)} = \frac{Ax+B}{1-9x^2} + \frac{Cx+D}{1-16x^2}.$$

Then $12x^2 + 1 \equiv (Ax + B)(1 - 16x^2) + (Cx + D)(1 - 9x^2)$. Substituting $x = 1/3, -1/3, 1/4, -1/4$ gives respectively

$$\frac{A}{3} + B = -3, \quad -\frac{A}{3} + B = -3, \quad \frac{C}{4} + D = 4, \quad -\frac{C}{4} + D = 4,$$

so $A = C = 0$, $B = -3$, $D = 4$. Thus

$$\begin{aligned}\int \frac{12x^2+1}{(1-9x^2)(1-16x^2)}\,dx &= \int \left(-\frac{3}{1-(3x)^2} + \frac{4}{1-(4x)^2}\right) dx \\ &= -\operatorname{artanh}(3x) + \operatorname{artanh}(4x) + c.\end{aligned}$$

14 b)

Write

$$\frac{10x^2+47}{(1-4x^2)(25-x^2)} = \frac{Ax+B}{1-4x^2} + \frac{Cx+D}{25-x^2}.$$

Then $10x^2 + 47 \equiv (Ax + B)(25 - x^2) + (Cx + D)(1 - 4x^2)$. Substituting $x = 5, -5, 1/2, -1/2$ gives

$$5C + D = -3, \quad -5C + D = -3, \quad A/2 + B = 2, \quad -A/2 + B = 2.$$

Thus $A = C = 0$, $B = 2$ and $D = -3$. Therefore

$$\begin{aligned}\int \frac{10x^2+47}{(1-4x^2)(25-x^2)}\,dx &= \int \left(\frac{2}{1-(2x)^2} - \frac{3}{25-x^2}\right) dx \\ &= \operatorname{artanh}(2x) - \frac{3}{5}\operatorname{artanh}\left(\frac{x}{5}\right) + c.\end{aligned}$$

14 c)

Because the numerator and denominator both have degree four, first divide out the constant term $E = 108/36 = 3$:

$$\frac{108x^4+12x^3-237x^2-27x+25}{(1-9x^2)(9-4x^2)} = 3 + \frac{Ax+B}{1-9x^2} + \frac{Cx+D}{9-4x^2}.$$

Then

$$108x^4 + 12x^3 - 237x^2 - 27x + 25 = 3(1-9x^2)(9-4x^2) + (Ax+B)(9-4x^2) + (Cx+D)(1-9x^2).$$

Substituting $x = 1/3, -1/3, 3/2, -3/2$ gives $A = -3$, $B = 0$, $C = 0$ and $D = -2$. Hence

$$f(x) = 3 - \frac{3x}{1-9x^2} - \frac{2}{9-4x^2}.$$

Therefore

$$\begin{aligned}\int f(x)\,dx &= 3x + \frac{1}{6}\ln|1-9x^2| - \frac{2}{9}\int \frac{1}{1-(2x/3)^2}\,dx \\ &= 3x + \frac{1}{6}\ln|1-9x^2| - \frac{1}{3}\operatorname{artanh}\left(\frac{2}{3}x\right) + c.\end{aligned}$$

15)

$$\begin{aligned}\int \frac{1-2x}{1-x^2}\,dx &= \int \left(\frac{1}{1-x^2} - \frac{2x}{1-x^2}\right) dx \\ &= \operatorname{artanh} x + \ln|1-x^2| + c.\end{aligned}$$

16 a)

$$\begin{aligned}\int \frac{6x+1}{1-9x^2}\,dx &= \int \left(\frac{6x}{1-9x^2} + \frac{1}{1-(3x)^2}\right) dx \\ &= -\frac{1}{3}\ln|1-9x^2| + \frac{1}{3}\operatorname{artanh}(3x) + c.\end{aligned}$$

16 b)

$$\begin{aligned}\int \frac{2-8x}{4-x^2}\,dx &= \int \left(\frac{1}{2(1-(x/2)^2)} - \frac{8x}{4-x^2}\right) dx \\ &= \operatorname{artanh}\left(\frac{x}{2}\right) + 4\ln|4-x^2| + c.\end{aligned}$$

16 c)

$$\begin{aligned}\int \frac{8-3x}{16-9x^2}\,dx &= \int \left(\frac{1}{2(1-(3x/4)^2)} - \frac{3x}{16-9x^2}\right) dx \\ &= \frac{2}{3}\operatorname{artanh}\left(\frac{3}{4}x\right) + \frac{1}{6}\ln|16-9x^2| + c.\end{aligned}$$

16 d)

$$\begin{aligned}\int \frac{10x-4}{25-100x^2}\,dx &= \int \left(\frac{10x}{25-100x^2} - \frac{4}{25(1-(2x)^2)}\right) dx \\ &= -\frac{1}{20}\ln|25-100x^2| - \frac{2}{25}\operatorname{artanh}(2x) + c.\end{aligned}$$

17)

$$\begin{aligned}\int \frac{1+x}{\sqrt{x^2-1}}\,dx &= \int \left(\frac{1}{\sqrt{x^2-1}} + \frac{x}{\sqrt{x^2-1}}\right) dx \\ &= \operatorname{arcosh} x + \sqrt{x^2-1} + c.\end{aligned}$$

18)

$$\begin{aligned}\int \frac{4x+1}{\sqrt{x^2+1}}\,dx &= \int \left(\frac{4x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}}\right) dx \\ &= 4\sqrt{x^2+1} + \operatorname{arsinh} x + c.\end{aligned}$$

19 a)

$$\begin{aligned}\int \frac{2x-1}{\sqrt{4x^2-1}}\,dx &= \int \frac{2x}{\sqrt{4x^2-1}}\,dx - \int \frac{1}{\sqrt{(2x)^2-1}}\,dx \\ &= \frac{1}{2}\sqrt{4x^2-1} - \frac{1}{2}\operatorname{arcosh}(2x) + c.\end{aligned}$$

19 b)

$$\begin{aligned}\int \frac{1-3x}{\sqrt{9x^2+1}}\,dx &= \int \frac{1}{\sqrt{(3x)^2+1}}\,dx - \int \frac{3x}{\sqrt{9x^2+1}}\,dx \\ &= \frac{1}{3}\operatorname{arsinh}(3x) - \frac{1}{3}\sqrt{9x^2+1} + c.\end{aligned}$$

19 c)

$$\begin{aligned}\int \frac{4+6x}{\sqrt{9x^2-4}}\,dx &= 4\int \frac{1}{\sqrt{(3x)^2-2^2}}\,dx + 6\int \frac{x}{\sqrt{9x^2-4}}\,dx \\ &= \frac{4}{3}\operatorname{arcosh}\left(\frac{3}{2}x\right) + \frac{2}{3}\sqrt{9x^2-4} + c.\end{aligned}$$

19 d)

$$\begin{aligned}\int \frac{8x-3}{\sqrt{4x^2+36}}\,dx &= 8\int \frac{x}{\sqrt{4x^2+36}}\,dx - 3\int \frac{1}{\sqrt{4x^2+36}}\,dx \\ &= 2\sqrt{4x^2+36} - \frac{3}{2}\operatorname{arsinh}\left(\frac{x}{3}\right) + c.\end{aligned}$$

20 a)

Let $x = \sinh u$, so $dx = \cosh u\, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{x^2+1}}\, dx &= \int \frac{\cosh u}{\sqrt{\sinh^2 u + 1}}\, du = \int \frac{\cosh u}{\sqrt{\cosh^2 u}}\, du \\ &= \int 1\, du = u + c = \operatorname{arsinh} x + c.\end{aligned}$$

20 b)

Let $x = \cosh u$, so $dx = \sinh u\, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{x^2-1}}\, dx &= \int \frac{\sinh u}{\sqrt{\cosh^2 u - 1}}\, du = \int \frac{\sinh u}{\sqrt{\sinh^2 u}}\, du \\ &= \int 1\, du = u + c = \operatorname{arcosh} x + c.\end{aligned}$$

20 c)

Let $x = \tanh u$, so $dx = \operatorname{sech}^2 u\, du$. Then

$$\int \frac{1}{1-x^2}\, dx = \int \frac{\operatorname{sech}^2 u}{1 - \tanh^2 u}\, du = \int 1\, du = u + c = \operatorname{artanh} x + c.$$

20 d)

Let $2x = \sinh u$, so $dx = \frac{1}{2}\cosh u\, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{4x^2+1}}\, dx &= \frac{1}{2}\int \frac{\cosh u}{\sqrt{\sinh^2 u + 1}}\, du = \frac{1}{2}\int 1\, du \\ &= \frac{1}{2}\operatorname{arsinh}(2x) + c.\end{aligned}$$

20 e)

Let $x = 3\cosh u$, so $dx = 3\sinh u\, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{x^2-9}}\, dx &= \int \frac{3\sinh u}{\sqrt{9\cosh^2 u - 9}}\, du = \int 1\, du \\ &= \operatorname{arcosh}\left(\frac{x}{3}\right) + c.\end{aligned}$$

20 f)

Let $3x = \tanh u$, so $dx = \frac{1}{3}\operatorname{sech}^2 u\, du$. Then

$$\begin{aligned}\int \frac{1}{1-9x^2}\, dx &= \frac{1}{3}\int \frac{\operatorname{sech}^2 u}{1 - \tanh^2 u}\, du = \frac{1}{3}u + c \\ &= \frac{1}{3}\operatorname{artanh}(3x) + c.\end{aligned}$$

20 g)

Let $x = 5\sinh u$, so $dx = 5\cosh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{x^2+25}}\,dx &= \int \frac{5\cosh u}{\sqrt{25\sinh^2 u + 25}}\,du = \int 1\,du \\ &= \operatorname{arsinh}\left(\frac{x}{5}\right) + c.\end{aligned}$$

20 h)

Let $6x = \cosh u$, so $dx = \frac{1}{6}\sinh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{36x^2-1}}\,dx &= \frac{1}{6}\int \frac{\sinh u}{\sqrt{\cosh^2 u - 1}}\,du = \frac{1}{6}\int 1\,du \\ &= \frac{1}{6}\operatorname{arcosh}(6x) + c.\end{aligned}$$

20 i)

Let $x = 10\tanh u$, so $dx = 10\operatorname{sech}^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{100-x^2}\,dx &= \frac{1}{10}\int \frac{10\operatorname{sech}^2 u}{1-\tanh^2 u}\,du = \frac{1}{10}u + c \\ &= \frac{1}{10}\operatorname{artanh}\left(\frac{x}{10}\right) + c.\end{aligned}$$

20 j)

Let $7x = \sinh u$, so $dx = \frac{1}{7}\cosh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{49x^2+1}}\,dx &= \frac{1}{7}\int \frac{\cosh u}{\sqrt{\sinh^2 u + 1}}\,du = \frac{1}{7}u + c \\ &= \frac{1}{7}\operatorname{arsinh}(7x) + c.\end{aligned}$$

20 k)

Let $x = 8\cosh u$, so $dx = 8\sinh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{x^2-64}}\,dx &= \int \frac{8\sinh u}{\sqrt{64\cosh^2 u - 64}}\,du = \int 1\,du \\ &= \operatorname{arcosh}\left(\frac{x}{8}\right) + c.\end{aligned}$$

20 l)

Let $12x = \tanh u$, so $dx = \frac{1}{12}\operatorname{sech}^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{1-144x^2}\,dx &= \frac{1}{12}\int \frac{\operatorname{sech}^2 u}{1-\tanh^2 u}\,du = \frac{1}{12}u + c \\ &= \frac{1}{12}\operatorname{artanh}(12x) + c.\end{aligned}$$

20 m)

Let $2x = 3\sinh u$, so $dx = \frac{3}{2}\cosh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{4x^2+9}}\,dx &= \frac{3}{2}\int \frac{\cosh u}{\sqrt{9\sinh^2 u+9}}\,du = \frac{1}{2}\int 1\,du \\ &= \frac{1}{2}\operatorname{arsinh}\left(\frac{2x}{3}\right)+c.\end{aligned}$$

20 n)

Let $3x = 4\cosh u$, so $dx = \frac{4}{3}\sinh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{9x^2-16}}\,dx &= \frac{4}{3}\int \frac{\sinh u}{\sqrt{16\cosh^2 u-16}}\,du = \frac{1}{3}\int 1\,du \\ &= \frac{1}{3}\operatorname{arcosh}\left(\frac{3x}{4}\right)+c.\end{aligned}$$

20 o)

Let $5x = 2\tanh u$, so $dx = \frac{2}{5}\operatorname{sech}^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{4-25x^2}\,dx &= \frac{1}{10}\int \frac{\operatorname{sech}^2 u}{1-\tanh^2 u}\,du = \frac{1}{10}u+c \\ &= \frac{1}{10}\operatorname{artanh}\left(\frac{5}{2}x\right)+c.\end{aligned}$$

20 p)

Let $6x = 5\sinh u$, so $dx = \frac{5}{6}\cosh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{36x^2+25}}\,dx &= \frac{1}{6}\int \frac{\cosh u}{\sqrt{\sinh^2 u+1}}\,du = \frac{1}{6}u+c \\ &= \frac{1}{6}\operatorname{arsinh}\left(\frac{6}{5}x\right)+c.\end{aligned}$$

20 q)

Let $20x = 3\cosh u$, so $dx = \frac{3}{20}\sinh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{400x^2-9}}\,dx &= \frac{1}{20}\int \frac{\sinh u}{\sqrt{\cosh^2 u-1}}\,du = \frac{1}{20}u+c \\ &= \frac{1}{20}\operatorname{arcosh}\left(\frac{20x}{3}\right)+c.\end{aligned}$$

20 r)

Let $6x = 9\tanh u$, so $dx = \frac{3}{2}\operatorname{sech}^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{81-36x^2}\,dx &= \frac{1}{54}\int \frac{\operatorname{sech}^2 u}{1-\tanh^2 u}\,du = \frac{1}{54}u+c \\ &= \frac{1}{54}\operatorname{artanh}\left(\frac{2}{3}x\right)+c.\end{aligned}$$

20 s)

Let $\sqrt{2}x = 3\sinh u$, so $dx = \frac{3}{\sqrt{2}}\cosh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{2x^2+9}}\,dx &= \frac{1}{\sqrt{2}}\int \frac{\cosh u}{\sqrt{\sinh^2 u+1}}\,du = \frac{u}{\sqrt{2}}+c\\ &= \frac{1}{\sqrt{2}}\operatorname{arsinh}\left(\frac{\sqrt{2}x}{3}\right)+c.\end{aligned}$$

20 t)

Let $5x = \sqrt{3}\cosh u$, so $dx = \frac{\sqrt{3}}{5}\sinh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{25x^2-3}}\,dx &= \frac{1}{5}\int \frac{\sinh u}{\sqrt{\cosh^2 u-1}}\,du = \frac{1}{5}u+c\\ &= \frac{1}{5}\operatorname{arcosh}\left(\frac{5x}{\sqrt{3}}\right)+c.\end{aligned}$$

20 u)

Let $\sqrt{5}x = 3\tanh u$, so $dx = \frac{3}{\sqrt{5}}\operatorname{sech}^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{9-5x^2}\,dx &= \frac{1}{3\sqrt{5}}\int \frac{\operatorname{sech}^2 u}{1-\tanh^2 u}\,du = \frac{u}{3\sqrt{5}}+c\\ &= \frac{1}{3\sqrt{5}}\operatorname{artanh}\left(\frac{\sqrt{5}x}{3}\right)+c.\end{aligned}$$

20 v)

Let $\sqrt{3}x = 2\sqrt{2}\sinh u$, so $dx = \frac{2\sqrt{2}}{\sqrt{3}}\cosh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{3x^2+8}}\,dx &= \frac{1}{\sqrt{3}}\int \frac{\cosh u}{\sqrt{\sinh^2 u+1}}\,du = \frac{u}{\sqrt{3}}+c\\ &= \frac{1}{\sqrt{3}}\operatorname{arsinh}\left(\frac{\sqrt{3}x}{2\sqrt{2}}\right)+c.\end{aligned}$$

20 w)

Let $2\sqrt{3}x = \sqrt{15}\cosh u$, so $dx = \frac{\sqrt{15}}{2\sqrt{3}}\sinh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{12x^2-15}}\,dx &= \frac{1}{2\sqrt{3}}\int \frac{\sinh u}{\sqrt{\cosh^2 u-1}}\,du = \frac{u}{2\sqrt{3}}+c\\ &= \frac{1}{2\sqrt{3}}\operatorname{arcosh}\left(\frac{2x}{\sqrt{5}}\right)+c.\end{aligned}$$

20 x)

Let $\sqrt{7}x = 2\sqrt{6}\tanh u$, so $dx = \frac{2\sqrt{6}}{\sqrt{7}}\operatorname{sech}^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{24-7x^2}\,dx &= \frac{1}{2\sqrt{42}}\int \frac{\operatorname{sech}^2 u}{1-\tanh^2 u}\,du = \frac{u}{2\sqrt{42}}+c\\ &= \frac{1}{2\sqrt{42}}\operatorname{artanh}\left(\frac{\sqrt{7}x}{2\sqrt{6}}\right)+c.\end{aligned}$$

20 y)

Let $x = \sqrt{a/b}\tanh u$, so $dx = \sqrt{a/b}\operatorname{sech}^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{a-bx^2}\,dx &= \int \frac{\sqrt{a/b}\operatorname{sech}^2 u}{a(1-\tanh^2 u)}\,du = \frac{1}{\sqrt{ab}}u + c \\ &= \frac{1}{\sqrt{ab}}\operatorname{artanh}\left(\sqrt{\frac{b}{a}}x\right) + c.\end{aligned}$$

20 z)

Let $x = \sqrt{b/a}\sinh u$, so $dx = \sqrt{b/a}\cosh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{ax^2+b}}\,dx &= \frac{1}{\sqrt{a}}\int \frac{\cosh u}{\sqrt{\sinh^2 u+1}}\,du = \frac{1}{\sqrt{a}}u + c \\ &= \frac{1}{\sqrt{a}}\operatorname{arsinh}\left(\sqrt{\frac{a}{b}}x\right) + c.\end{aligned}$$

20 aa)

Let $x = \sqrt{b/a}\cosh u$, so $dx = \sqrt{b/a}\sinh u\,du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{ax^2-b}}\,dx &= \frac{1}{\sqrt{a}}\int \frac{\sinh u}{\sqrt{\cosh^2 u-1}}\,du = \frac{1}{\sqrt{a}}u + c \\ &= \frac{1}{\sqrt{a}}\operatorname{arcosh}\left(\sqrt{\frac{a}{b}}x\right) + c.\end{aligned}$$

21 a)

Let $x = \sinh u$, so $dx = \cosh u\,du$. Using $\cosh^2 u = (\cosh 2u + 1)/2$,

$$\begin{aligned}\int \sqrt{x^2+1}\,dx &= \int \cosh^2 u\,du = \frac{1}{4}\sinh 2u + \frac{1}{2}u + c \\ &= \frac{1}{2}\sinh u\cosh u + \frac{1}{2}u + c = \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\operatorname{arsinh} x + c.\end{aligned}$$

21 b)

Let $x = \sinh u$, so $dx = \cosh u\,du$. Then

$$\int (x^2+1)^{3/2}\,dx = \int \cosh^4 u\,du.$$

Since $\cosh^2 u = (\cosh 2u + 1)/2$,

$$\begin{aligned}\cosh^4 u &= \frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}, \\ \int \cosh^4 u\,du &= \frac{1}{32}\sinh 4u + \frac{1}{4}\sinh 2u + \frac{3}{8}u + c \\ &= \frac{1}{8}x\sqrt{x^2+1}(2x^2+5) + \frac{3}{8}\operatorname{arsinh} x + c.\end{aligned}$$

22 a)

Let $x = \cosh u$, so $dx = \sinh u\, du$. Then

$$\begin{aligned}\int \sqrt{x^2-1}\, dx &= \int \sinh^2 u\, du = \frac{1}{4}\sinh 2u - \frac{1}{2}u + c \\ &= \frac{1}{2}x\sqrt{x^2-1} - \frac{1}{2}\operatorname{arcosh} x + c.\end{aligned}$$

22 b)

Let $x = \cosh u$, so $dx = \sinh u\, du$. Then

$$\int (x^2-1)^{3/2}\, dx = \int \sinh^4 u\, du.$$

Using $\sinh^2 u = (\cosh 2u - 1)/2$,

$$\begin{aligned}\sinh^4 u &= \frac{1}{8}\cosh 4u - \frac{1}{2}\cosh 2u + \frac{3}{8}, \\ \int \sinh^4 u\, du &= \frac{1}{32}\sinh 4u - \frac{1}{4}\sinh 2u + \frac{3}{8}u + c \\ &= \frac{1}{8}x\sqrt{x^2-1}(2x^2-5) + \frac{3}{8}\operatorname{arcosh} x + c.\end{aligned}$$

23 a)

Let $x = \tanh u$, so $dx = \operatorname{sech}^2 u\, du$ and $1 - x^2 = \operatorname{sech}^2 u$. Then

$$\begin{aligned}\int \frac{1}{(1-x^2)^2}\, dx &= \int \cosh^2 u\, du = \frac{1}{4}\sinh 2u + \frac{1}{2}u + c \\ &= \frac{x}{2(1-x^2)} + \frac{1}{2}\operatorname{artanh} x + c.\end{aligned}$$

23 b)

Let $3x = \tanh u$, so $dx = \frac{1}{3}\operatorname{sech}^2 u\, du$. Then

$$\begin{aligned}\int \frac{1}{(1-9x^2)^2}\, dx &= \frac{1}{3}\int \cosh^2 u\, du = \frac{1}{12}\sinh 2u + \frac{1}{6}u + c \\ &= \frac{x}{2(1-9x^2)} + \frac{1}{6}\operatorname{artanh}(3x) + c.\end{aligned}$$

23 c)

Let $x = \tanh u$, so $dx = \operatorname{sech}^2 u\, du$. Then

$$\begin{aligned}\int \frac{1}{(1-x^2)^3}\, dx &= \int \cosh^4 u\, du \\ &= \frac{1}{32}\sinh 4u + \frac{1}{4}\sinh 2u + \frac{3}{8}u + c \\ &= \frac{5x - 3x^3}{8(1-x^2)^2} + \frac{3}{8}\operatorname{artanh} x + c.\end{aligned}$$

23 d)

Let $x = 2\tanh u$, so $dx = 2\operatorname{sech}^2 u\,du$. Then

$$\begin{aligned}\int \frac{1}{(4-x^2)^3}\,dx &= \frac{1}{32}\int \cosh^4 u\,du \\ &= \frac{1}{1024}\sinh 4u + \frac{1}{128}\sinh 2u + \frac{3}{256}u + c \\ &= \frac{20x - 3x^3}{128(4-x^2)^2} + \frac{3}{256}\operatorname{artanh}\left(\frac{x}{2}\right) + c.\end{aligned}$$

24 a)

Let $5x = \sinh u$, so $dx = \frac{1}{5}\cosh u\,du$. Then

$$\begin{aligned}\int \sqrt{25x^2+1}\,dx &= \frac{1}{5}\int \cosh^2 u\,du = \frac{1}{20}\sinh 2u + \frac{1}{10}u + c \\ &= \frac{1}{2}x\sqrt{25x^2+1} + \frac{1}{10}\operatorname{arsinh}(5x) + c.\end{aligned}$$

24 b)

Let $x = 3\sinh u$, so $dx = 3\cosh u\,du$. Then

$$\begin{aligned}\int (x^2+9)^{3/2}\,dx &= 81\int \cosh^4 u\,du \\ &= \frac{81}{32}\sinh 4u + \frac{81}{4}\sinh 2u + \frac{243}{8}u + c \\ &= \frac{1}{8}x\sqrt{x^2+9}(2x^2+45) + \frac{243}{8}\operatorname{arsinh}\left(\frac{x}{3}\right) + c.\end{aligned}$$

24 c)

Let $x = 6\cosh u$, so $dx = 6\sinh u\,du$. Then

$$\begin{aligned}\int \sqrt{x^2-36}\,dx &= 36\int \sinh^2 u\,du = 9\sinh 2u - 18u + c \\ &= \frac{1}{2}x\sqrt{x^2-36} - 18\operatorname{arcosh}\left(\frac{x}{6}\right) + c.\end{aligned}$$

24 d)

Let $9x = \cosh u$, so $dx = \frac{1}{9}\sinh u\,du$. Then

$$\begin{aligned}\int (81x^2-1)^{3/2}\,dx &= \frac{1}{9}\int \sinh^4 u\,du \\ &= \frac{1}{288}\sinh 4u - \frac{1}{36}\sinh 2u + \frac{1}{24}u + c \\ &= \frac{1}{8}x(162x^2-5)\sqrt{81x^2-1} + \frac{1}{24}\operatorname{arcosh}(9x) + c.\end{aligned}$$

25 a)

$$x^2 + 4x + 3 = (x+2)^2 - 4 + 3 = (x+2)^2 - 1.$$

25 b)

$$x^2 - \frac{10}{3}x + \frac{34}{9} = \left(x - \frac{5}{3}\right)^2 - \frac{25}{9} + \frac{34}{9} = \left(x - \frac{5}{3}\right)^2 + 1.$$

25 c)

$$4x^2 + 24x + 37 = 4(x^2 + 6x) + 37 = 4((x+3)^2 - 9) + 37 = 4(x+3)^2 + 1.$$

25 d)

$$9x^2 - 18x + 8 = 9(x^2 - 2x) + 8 = 9((x-1)^2 - 1) + 8 = 9(x-1)^2 - 1.$$

25 e)

$$x^2 + 6x + 18 = (x+3)^2 - 9 + 18 = (x+3)^2 + 9.$$

25 f)

$$9x^2 - 36x + 32 = 9(x^2 - 4x) + 32 = 9((x-2)^2 - 4) + 32 = 9(x-2)^2 - 4.$$

26 a)

$$x^2 + 20x + 101 = (x+10)^2 - 100 + 101 = (x+10)^2 + 1.$$

26 b)

Let $x + 10 = \sinh u$, so $dx = \cosh u \, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{x^2 + 20x + 101}} \, dx = \int \frac{\cosh u}{\sqrt{\sinh^2 u + 1}} \, du &= \int 1 \, du \\ &= \operatorname{arsinh}(x+10) + c.\end{aligned}$$

27 a)

$$x^2 - 14x + 48 = (x-7)^2 - 49 + 48 = (x-7)^2 - 1.$$

27 b)

Let $x - 7 = \cosh u$, so $dx = \sinh u \, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{x^2 - 14x + 48}}\, dx &= \int \frac{\sinh u}{\sqrt{\cosh^2 u - 1}}\, du = \int 1 \, du \\ &= \operatorname{arcosh}(x-7) + c.\end{aligned}$$

28 a)

$$\int \frac{1}{\sqrt{x^2 - 10x + 26}}\, dx = \int \frac{1}{\sqrt{(x-5)^2 + 1}}\, dx = \operatorname{arsinh}(x-5) + c.$$

28 b)

$$\int \frac{1}{\sqrt{x^2 + \frac{2}{3}x - \frac{8}{9}}}\, dx = \int \frac{1}{\sqrt{(x+\frac{1}{3})^2 - 1}}\, dx = \operatorname{arcosh}\left(x + \frac{1}{3}\right) + c.$$

29)

The interval length is 4/5. Completing the square,

$$x^2 - \frac{4}{5}x + \frac{29}{25} = \left(x - \frac{2}{5}\right)^2 + 1.$$

Therefore the mean value is

$$\begin{aligned}\frac{1}{4/5}\int_{-2/5}^{2/5} f(x)\, dx &= \frac{5}{4}\left[\operatorname{arsinh}\left(x - \frac{2}{5}\right)\right]_{-2/5}^{2/5} \\ &= \frac{5}{4}\left(\operatorname{arsinh} 0 - \operatorname{arsinh}\left(-\frac{4}{5}\right)\right) \\ &= -\frac{5}{4}\ln\left(\frac{\sqrt{41} - 4}{5}\right).\end{aligned}$$

30 a)

$$4x^2 - 40x + 101 = 4(x^2 - 10x) + 101 = 4((x-5)^2 - 25) + 101 = 4(x-5)^2 + 1.$$

30 b)

Let $2x - 10 = \sinh u$, so $dx = \frac{1}{2}\cosh u\, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{4x^2 - 40x + 101}}\, dx &= \frac{1}{2}\int \frac{\cosh u}{\sqrt{\sinh^2 u + 1}}\, du = \frac{1}{2}u + c \\ &= \frac{1}{2}\operatorname{arsinh}(2x - 10) + c.\end{aligned}$$

31 a)

$$16x^2 + 96x + 143 = 16(x^2 + 6x) + 143 = 16((x+3)^2 - 9) + 143 = 16(x+3)^2 - 1.$$

31 b)

Let $4x + 12 = \cosh u$, so $dx = \frac{1}{4}\sinh u\, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{16x^2 + 96x + 143}}\, dx &= \frac{1}{4}\int \frac{\sinh u}{\sqrt{\cosh^2 u - 1}}\, du = \frac{1}{4}u + c \\ &= \frac{1}{4}\operatorname{arcosh}(4x + 12) + c.\end{aligned}$$

32 a)

$$9x^2 - 144x + 575 = 9(x-8)^2 - 1 = (3x - 24)^2 - 1,$$

so

$$\int \frac{1}{\sqrt{9x^2 - 144x + 575}}\, dx = \frac{1}{3}\operatorname{arcosh}(3x - 24) + c.$$

32 b)

$$2x^2 + 12x + 19 = 2(x+3)^2 + 1 = (\sqrt{2}x + 3\sqrt{2})^2 + 1,$$

so

$$\int \frac{1}{\sqrt{2x^2 + 12x + 19}}\, dx = \frac{1}{\sqrt{2}}\operatorname{arsinh}(\sqrt{2}x + 3\sqrt{2}) + c.$$

33 a)

$$4x^2 - 56x + 205 = 4(x^2 - 14x) + 205 = 4((x-7)^2 - 49) + 205 = 4(x-7)^2 + 9.$$

33 b)

Let $\frac{2}{3}(x - 7) = \sinh u$, so $dx = \frac{3}{2}\cosh u\, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{4x^2 - 56x + 205}}\, dx &= \frac{1}{3}\int \frac{\frac{3}{2}\cosh u}{\sqrt{\sinh^2 u + 1}}\, du = \frac{1}{2}u + c \\ &= \frac{1}{2}\operatorname{arsinh}\left(\frac{2}{3}(x-7)\right) + c.\end{aligned}$$

34 a)

$$3x^2 + 6x - 1 = 3(x+1)^2 - 4.$$

34 b)

Let $\frac{\sqrt{3}}{2}(x+1) = \cosh u$, so $dx = \frac{2}{\sqrt{3}} \sinh u \, du$. Then

$$\begin{aligned}\int \frac{1}{\sqrt{3x^2+6x-1}}\,dx &= \frac{2}{\sqrt{3}} \int \frac{\sinh u}{\sqrt{\cosh^2 u - 1}}\,du = \frac{2}{\sqrt{3}}u + c \\ &= \frac{2}{\sqrt{3}} \operatorname{arcosh}\left(\frac{\sqrt{3}}{2}(x+1)\right) + c.\end{aligned}$$

35 a)

$$25x^2 - 100x + 91 = 25(x-2)^2 - 9 = 9\left(\left(\frac{5}{3}(x-2)\right)^2 - 1\right),$$

so

$$\begin{aligned}\int \frac{1}{\sqrt{25x^2-100x+91}}\,dx &= \frac{1}{3}\int \frac{1}{\sqrt{(\frac{5}{3}(x-2))^2 - 1}}\,dx \\ &= \frac{1}{5} \operatorname{arcosh}\left(\frac{5}{3}x - \frac{10}{3}\right) + c.\end{aligned}$$

35 b)

$$5x^2 + 10\sqrt{5}x + 29 = 5(x+\sqrt{5})^2 + 4 = 4\left(\left(\frac{\sqrt{5}}{2}x + \frac{5}{2}\right)^2 + 1\right),$$

so

$$\begin{aligned}\int \frac{1}{\sqrt{5x^2+10\sqrt{5}x+29}}\,dx &= \frac{1}{2}\int \frac{1}{\sqrt{(\frac{\sqrt{5}}{2}x + \frac{5}{2})^2 + 1}}\,dx \\ &= \frac{\sqrt{5}}{5} \operatorname{arsinh}\left(\frac{\sqrt{5}}{2}x + \frac{5}{2}\right) + c.\end{aligned}$$

36)

$$4x^2 + 8x + 13 = 4(x+1)^2 + 9 = 9\left(\left(\frac{2}{3}(x+1)\right)^2 + 1\right),$$

$$\begin{aligned}\int_{-1}^{1} \frac{1}{\sqrt{4x^2+8x+13}}\,dx &= \frac{1}{2}\left[\operatorname{arsinh}\left(\frac{2}{3}(x+1)\right)\right]_{-1}^{1} \\ &= \frac{1}{2}\operatorname{arsinh}\frac{4}{3} = \frac{1}{2}\ln 3.\end{aligned}$$

Thus $k = 3$.

37 a)

$$-3 - 4x - x^2 = 1 - (x+2)^2.$$

37 b)

$$-24 + 10x - x^2 = 1 - (x-5)^2.$$

37 c)

$$-15 - 16x - 4x^2 = 1 - 4(x+2)^2.$$

37 d)

$$-143 + 72x - 9x^2 = 1 - 9(x-4)^2.$$

37 e)

$$-45 - 14x - x^2 = 4 - (x+7)^2.$$

37 f)

$$\begin{aligned} 8 + 4x - 4x^2 &= 8 - 4(x^2 - x) = 8 - 4\left((x - \tfrac{1}{2})^2 - \tfrac{1}{4}\right) \\ &= 9 - 4(x - \tfrac{1}{2})^2. \end{aligned}$$

38 a)

$$-63 + 16x - x^2 = 1 - (x-8)^2.$$

38 b)

Completing the square gives $-63 + 16x - x^2 = 1 - (x-8)^2$. With $x - 8 = \tanh u$, $dx = \operatorname{sech}^2 u \, du$ and $1 - (x-8)^2 = \operatorname{sech}^2 u$. Thus

$$\int \frac{1}{-63 + 16x - x^2}\, dx = \int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\, du = u + c = \operatorname{artanh}(x-8) + c.$$

39 a)

$$-99 - 20x - x^2 = 1 - (x+10)^2,$$

$$\int \frac{1}{-99 - 20x - x^2}\, dx = \operatorname{artanh}(x+10) + c.$$

39 b)

$$-168 + 26x - x^2 = 1 - (x - 13)^2,$$
$$\int \frac{1}{-168 + 26x - x^2}\, dx = \operatorname{artanh}(x + 13) + c.$$

40 a)

$$-35 + 24x - 4x^2 = 1 - 4(x - 3)^2.$$

40 b)

With $2x - 6 = \tanh u$, $dx = \frac{1}{2}\operatorname{sech}^2 u\, du$ and $1 - (2x - 6)^2 = \operatorname{sech}^2 u$. Therefore

$$\int \frac{1}{-35 + 24x - 4x^2}\, dx = \frac{1}{2}\int 1\, du = \frac{1}{2}\operatorname{artanh}(2x - 6) + c.$$

41 a)

$$-224 - 90x - 9x^2 = 1 - (3x + 15)^2,$$
$$\int \frac{1}{-224 - 90x - 9x^2}\, dx = \frac{1}{3}\operatorname{artanh}(3x + 15) + c.$$

41 b)

$$-24 + 50x - 25x^2 = 1 - (5x - 5)^2,$$
$$\int \frac{1}{-24 + 50x - 25x^2}\, dx = \frac{1}{5}\operatorname{artanh}(5x - 5) + c.$$

42 a)

$$11 + 10x - x^2 = 36 - (x - 5)^2.$$

42 b)

With $(x - 5)/6 = \tanh u$, $dx = 6\operatorname{sech}^2 u\, du$ and $36 - (x - 5)^2 = 36\operatorname{sech}^2 u$. Thus

$$\int \frac{1}{11 + 10x - x^2}\, dx = \frac{1}{6}\int 1\, du = \frac{1}{6}\operatorname{artanh}\left(\frac{x - 5}{6}\right) + c.$$

43 a)

$$91 - 18x - 9x^2 = 100 - 9(x + 1)^2,$$
$$\int \frac{1}{91 - 18x - 9x^2}\, dx = \frac{1}{30}\operatorname{artanh}\left(\frac{3(x + 1)}{10}\right) + c.$$

43 b)

$$-20 - 24x - 4x^2 = 16 - 4(x+3)^2,$$
$$\int \frac{1}{-20 - 24x - 4x^2}\,dx = \frac{1}{8}\operatorname{artanh}\left(\frac{x+3}{2}\right) + c.$$

44)

The shaded region is under the curve from $x = 0$ to $x = 5$. Since

$$63 + 12x - 3x^2 = 3\left(25 - (x-2)^2\right),$$

its area is

$$\begin{aligned} A &= \int_0^5 \frac{5}{63 + 12x - 3x^2}\,dx = \frac{1}{3}\left[\operatorname{artanh}\left(\frac{x-2}{5}\right)\right]_0^5 \\ &= \frac{1}{3}\left(\operatorname{artanh}\frac{3}{5} + \operatorname{artanh}\frac{2}{5}\right) = \frac{1}{6}\ln\frac{28}{3}. \end{aligned}$$

45)

The volume of revolution is

$$\begin{aligned} V &= \pi\int_2^3 \left(\frac{2}{\sqrt{-16 + 16x - 3x^2}}\right)^2 dx \\ &= 4\pi\int_2^3 \frac{1}{-16 + 16x - 3x^2}\,dx = 4\pi\int_2^3 \frac{1}{-(3x-4)(x-4)}\,dx \\ &= \frac{\pi}{2}\left[\ln\left|\frac{3x-4}{x-4}\right|\right]_2^3 = \frac{\pi}{2}\ln 5. \end{aligned}$$

46)

Let $x = \cosh u$, so $dx = \sinh u\,du$. Then

$$\begin{aligned} \int \sqrt{x^2-1}\,dx &= \int \sinh^2 u\,du = \frac{1}{4}\sinh 2u - \frac{1}{2}u + c \\ &= \frac{1}{2}x\sqrt{x^2-1} - \frac{1}{2}\operatorname{arcosh} x + c. \end{aligned}$$

47)

Let $x = \sinh u$, so $dx = \cosh u\,du$. Then

$$\begin{aligned} \int \sqrt{x^2+1}\,dx &= \int \cosh^2 u\,du = \frac{1}{4}\sinh 2u + \frac{1}{2}u + c \\ &= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\operatorname{arsinh} x + c. \end{aligned}$$

48)

Let $x = \frac{1}{3}\cosh u$, so $dx = \frac{1}{3}\sinh u\,du$. Then

$$\begin{aligned} \int \sqrt{9x^2-1}\,dx &= \frac{1}{3}\int \sinh^2 u\,du = \frac{1}{12}\sinh 2u - \frac{1}{6}u + c \\ &= \frac{1}{2}x\sqrt{9x^2-1} - \frac{1}{6}\operatorname{arcosh}(3x) + c. \end{aligned}$$

49)

Let $x = 12\cosh u$, so $dx = 12\sinh u\,du$. The limits become $u = 0$ when $x = 12$, and $u = \operatorname{arcosh}(13/12) = \ln(3/2)$ when $x = 13$. Hence

$$\begin{aligned}\int_{12}^{13} \sqrt{x^2-144}\,dx &= 144\int_0^{\ln(3/2)} \sinh^2 u\,du \\ &= 72\left[\frac{1}{2}\sinh 2u - u\right]_0^{\ln(3/2)} \\ &= 18\left(\frac{9}{4}-\frac{4}{9}\right) - 72\ln\frac{3}{2} = \frac{65}{2} - 72\ln\frac{3}{2}.\end{aligned}$$

50)

Let $x = \frac{1}{5}\sinh u$. The limits are $u = 0$ and $u = \operatorname{arsinh} 10 = \ln(10+\sqrt{101})$. Thus

$$\begin{aligned}\int_0^2 \sqrt{25x^2+1}\,dx &= \frac{1}{5}\int_0^{\ln(10+\sqrt{101})} \cosh^2 u\,du \\ &= \frac{1}{10}\left[\frac{1}{2}\sinh 2u + u\right]_0^{\ln(10+\sqrt{101})} \\ &= \sqrt{101} + \frac{1}{10}\ln(10+\sqrt{101}).\end{aligned}$$

51 a)

Complete the square and use $x - 5 = \sinh u$:

$$\begin{aligned}\int \sqrt{x^2-10x+26}\,dx &= \int \sqrt{(x-5)^2+1}\,dx \\ &= \frac{1}{2}(x-5)\sqrt{(x-5)^2+1} + \frac{1}{2}\operatorname{arsinh}(x-5) + c.\end{aligned}$$

Equivalently, this is $\frac{1}{2}(x-5)\sqrt{(x-5)^2+1} + \frac{1}{2}\ln\left(x-5+\sqrt{(x-5)^2+1}\right) + c$.

51 b)

Complete the square and use $x + 3 = \cosh u$:

$$\begin{aligned}\int \sqrt{x^2+6x+8}\,dx &= \int \sqrt{(x+3)^2-1}\,dx \\ &= \frac{1}{2}(x+3)\sqrt{(x+3)^2-1} - \frac{1}{2}\operatorname{arcosh}(x+3) + c.\end{aligned}$$

Equivalently, the inverse hyperbolic term is $-\frac{1}{2}\ln\left(x+3+\sqrt{(x+3)^2-1}\right)$.

52)

Complete the square and let $x - 4 = \sinh u$. The limits $x = 3, 5$ become $u = \operatorname{arsinh}(-1)$ and $u = \operatorname{arsinh} 1$. Then

$$\begin{aligned}\int_3^5 \sqrt{x^2-8x+17}\,dx &= \int_{\operatorname{arsinh}(-1)}^{\operatorname{arsinh} 1} \cosh^2 u\,du \\ &= \left[\frac{1}{4}\sinh 2u + \frac{1}{2}u\right]_{\operatorname{arsinh}(-1)}^{\operatorname{arsinh} 1} \\ &= \sqrt{2} + \operatorname{arsinh} 1 = \sqrt{2} + \ln(1+\sqrt{2}).\end{aligned}$$

53)

Let $x = \cosh u$, so $dx = \sinh u\, du$. Then

$$\begin{aligned}\int \frac{x^2}{\sqrt{x^2-1}}\, dx &= \int \cosh^2 u\, du = \frac{1}{4}\sinh 2u + \frac{1}{2}u + c \\ &= \frac{1}{2}x\sqrt{x^2-1} + \frac{1}{2}\operatorname{arcosh} x + c.\end{aligned}$$

Equivalently, the final term is $\frac{1}{2}\ln(x + \sqrt{x^2-1})$.

54)

Let $x = \sinh u$, so $dx = \cosh u\, du$. Then

$$\begin{aligned}\int \frac{x^2}{\sqrt{x^2+1}}\, dx &= \int \sinh^2 u\, du = \frac{1}{4}\sinh 2u - \frac{1}{2}u + c \\ &= \frac{1}{2}x\sqrt{x^2+1} - \frac{1}{2}\operatorname{arsinh} x + c.\end{aligned}$$

Equivalently, the final term is $-\frac{1}{2}\ln(x + \sqrt{x^2+1})$.

55)

Let $x = \sinh u$, so $dx = \cosh u\, du$. Then

$$\begin{aligned}\int \frac{x^3}{\sqrt{x^2+1}}\, dx &= \int \sinh^3 u\, du = \int (\cosh^2 u - 1)\sinh u\, du \\ &= \frac{1}{3}\cosh^3 u - \cosh u + c \\ &= \frac{1}{3}(x^2+1)^{3/2} - \sqrt{x^2+1} + c.\end{aligned}$$

56)

Let $x = \sinh^2 u$, so $dx = 2\sinh u \cosh u\, du$. Then

$$\begin{aligned}\int \sqrt{\frac{x}{x+1}}\, dx &= \int \frac{\sinh u}{\cosh u}\, 2\sinh u \cosh u\, du = \int 2\sinh^2 u\, du \\ &= \frac{1}{2}\sinh 2u - u + c \\ &= \sqrt{x(x+1)} - \operatorname{arsinh}(\sqrt{x}) + c \\ &= \sqrt{x^2+x} - \ln\left(\sqrt{x} + \sqrt{x+1}\right) + c.\end{aligned}$$

57)

Let $x = \cosh u$, so $dx = \sinh u\, du$. Then

$$\begin{aligned}\int (x^2-1)^{3/2}\, dx &= \int \sinh^4 u\, du \\ &= \frac{1}{32}\sinh 4u - \frac{1}{4}\sinh 2u + \frac{3}{8}u + c \\ &= \frac{1}{8}x(2x^2-1)\sqrt{x^2-1} - \frac{1}{2}x\sqrt{x^2-1} + \frac{3}{8}\operatorname{arcosh} x + c \\ &= \frac{1}{8}x(2x^2-5)\sqrt{x^2-1} + \frac{3}{8}\operatorname{arcosh} x + c.\end{aligned}$$